Absolute Error and MAE

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Take the magnitude. Average the right quantity. Retain the unit.

Compare MAE,MSE and RMSE under matched cases and explicit loss priorities.

Keep one habit: Complete residual calculation → pictured unit conversion → same-case candidate/baseline contrast → independent new units.

MAIN LESSON · SECTION 1

Which error size does this loss summarize?

> Take the magnitude or square before averaging, and retain the correct unit.

Learning goals

I can calculate MAE MSE and **RMSE**.

I can explain how squaring changes sensitivity to large errors.

I can match the **loss** and **unit**s to the task.

Retrieve baseline and cancellation reasoning

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.
- Why must we take the absolute value of each residual before averaging to measure error size without cancellation?

End goal

By the end, I can explain **absolute error** and mae using a traced example and a stated limitation.

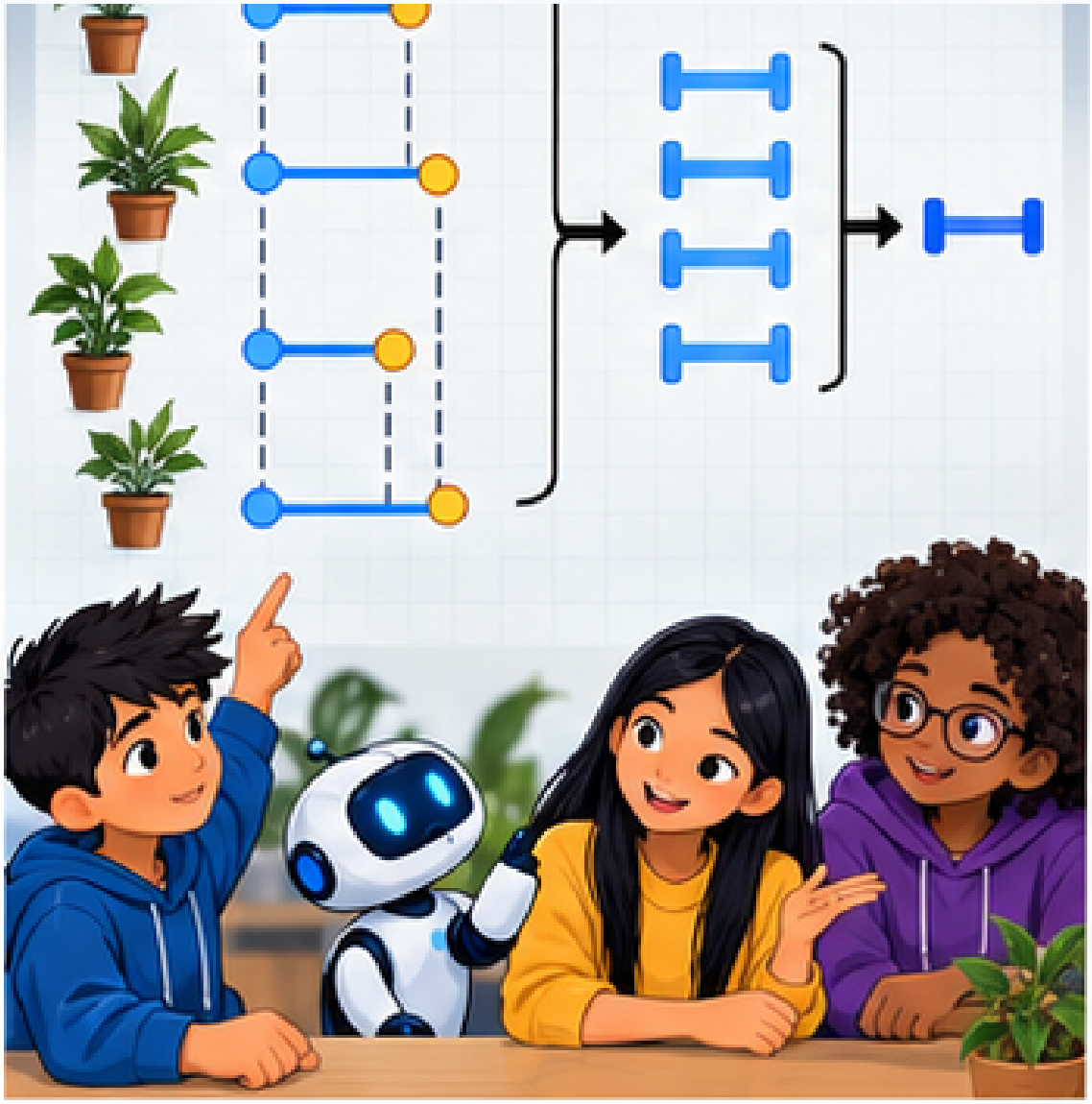
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Absolute error measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.
</details>



LEARNING TARGET Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE.
A student averages residuals and gets zero; is that MAE?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Define MAE, MSE and RMSE before comparing models

> The operation and unit tell the reader what a reported number means.

Complete loss definitions and fictional scope

A **loss** summarizes error magnitude without positive and negative **cancellation**. MAE averages absolute residuals. MSE averages squared residuals and penalizes large magnitudes more strongly.

RMSE takes the square root of MSE so it returns to the target **unit**. These are different questions, not interchangeable labels.

Use fictional classroom data throughout this investigation. Keep the task, **unit**s, reference labels and data split visible as you reason.

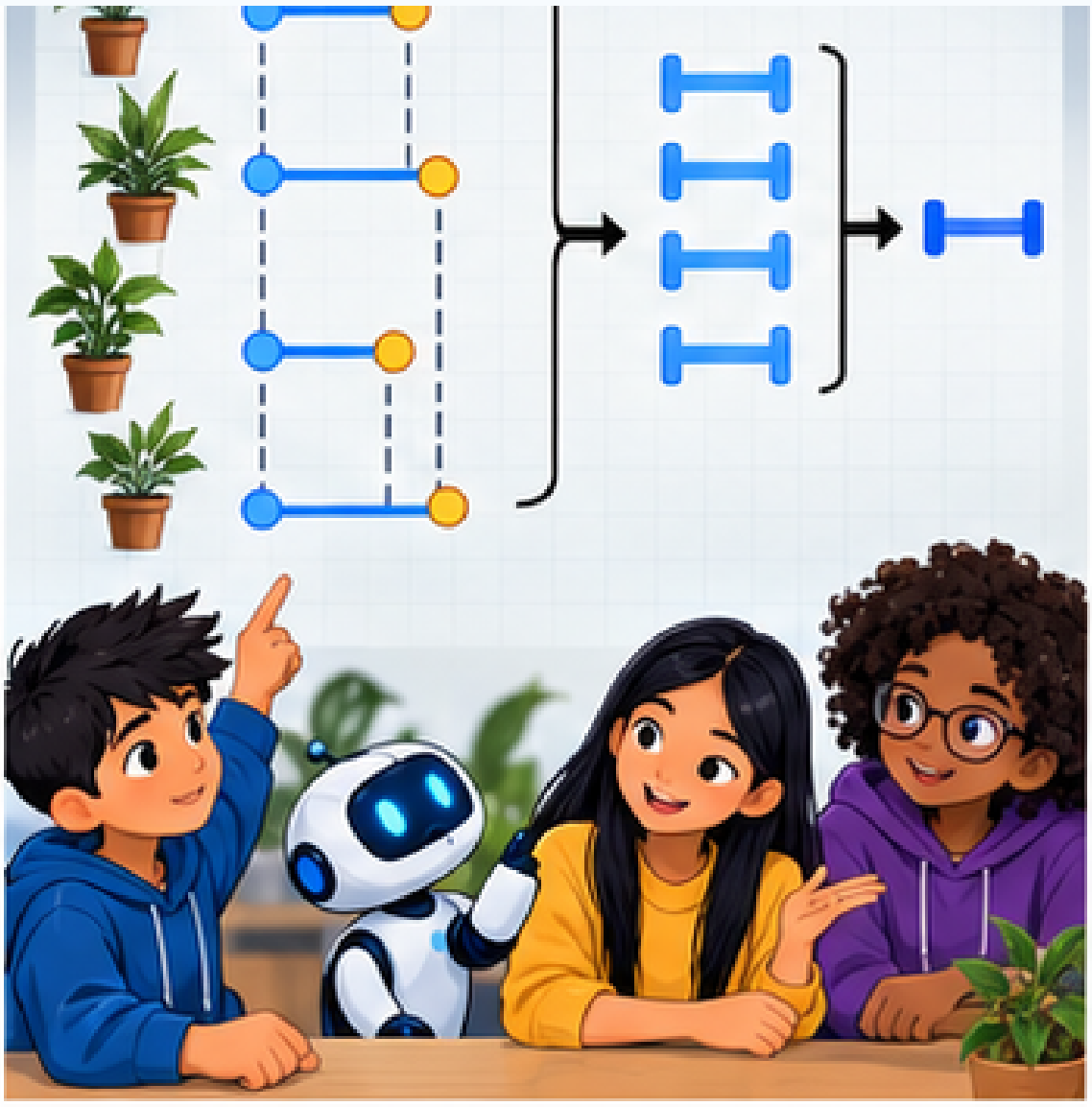
Earlier weeks supplied the model-building **process**; this week examines absolute and squared **loss**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **absolute error** and mae to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Trace magnitudes, squares and the square root in order

> The same residuals produce different summaries.

Complete worked residuals and calculations

Residuals [-1, 2,-3] cm have **absolute error**s [1, 2, 3], so MAE=6/3=2 cm. Squared errors [1, 4, 9] sum to 14, so MSE=14/3 square cm and **RMSE**=sqrt(14/3), about 2.16 cm. Keep the square and the average in the correct order.

Optional original calculation script

```
from math import sqrt
residuals = [-1, 2, -3]
mae = sum(abs(r) for r in residuals) / len(residuals)
mse = sum(r * r for r in residuals) / len(residuals)
print(mae, mse, sqrt(mse))
# About 2.0, 4.67, 2.16.
# MAE and RMSE: cm. MSE: square cm.
```

Predict and state the input assumptions

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

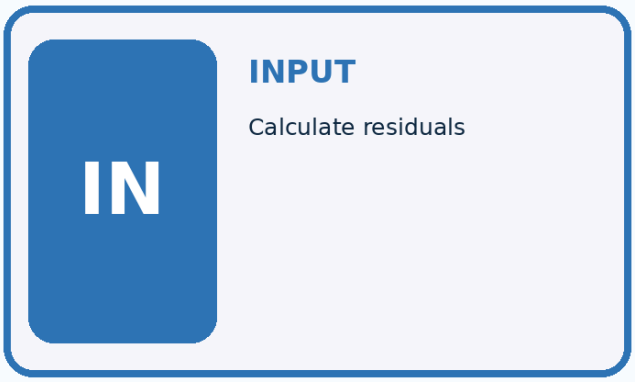
The original calculation assumes a nonempty list of finite residuals already paired and expressed in cm. Its numbers summarize supplied predictions; no model is fitted here. Average signed residual is a different statistic. MSE uses squared target units, while MAE and RMSE use the original unit. An empty or missing collection cannot be summarized as observed zero error.

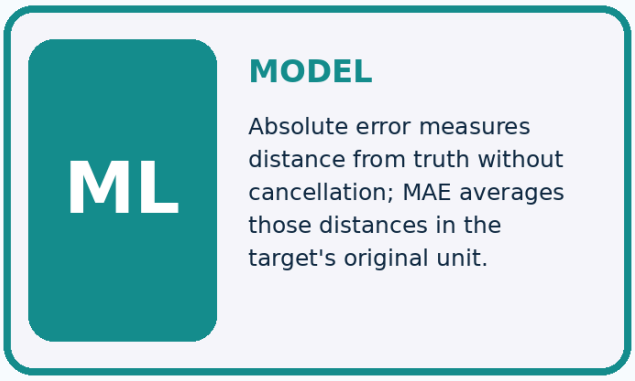
▼ **Earlier wording, alternatives and picture transcripts**

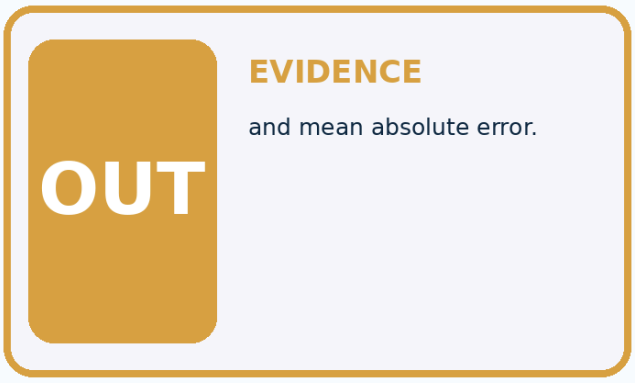
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Calculate residuals, **absolute error**s, and mean **absolute error**.

MODEL IDEA **Absolute error** measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17. Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

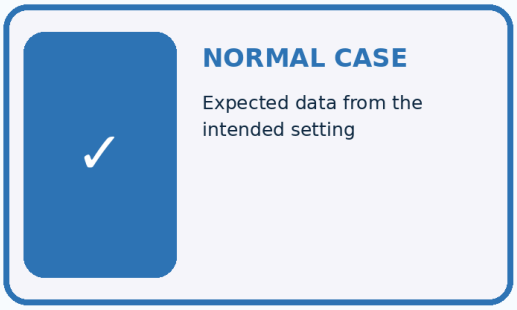
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

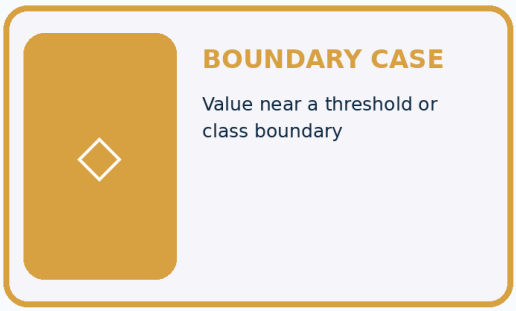
Compare two loss names with the operations they require

> A smaller number in different units or under a different loss is not automatically an improvement.

Four evaluation conditions



NORMAL Expected data and expected behavior.

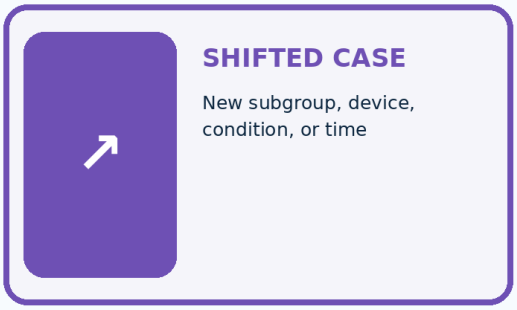


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Convert every residual to millimetres.

▼ **Check after predicting**

Prediction: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

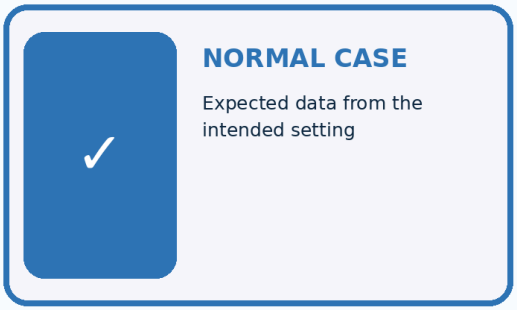
Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

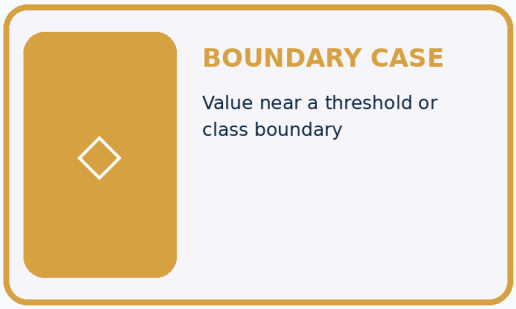
▼ **Earlier wording, alternatives and picture transcripts**

EVALUATION DESIGN

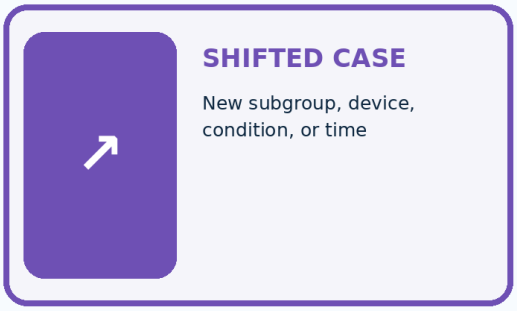
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Keep units beside every step in the picture

> Squaring the error also squares its unit.

Complete pictured residuals and scenario scope

Compare absolute and squared error without losing track of **unit**s.

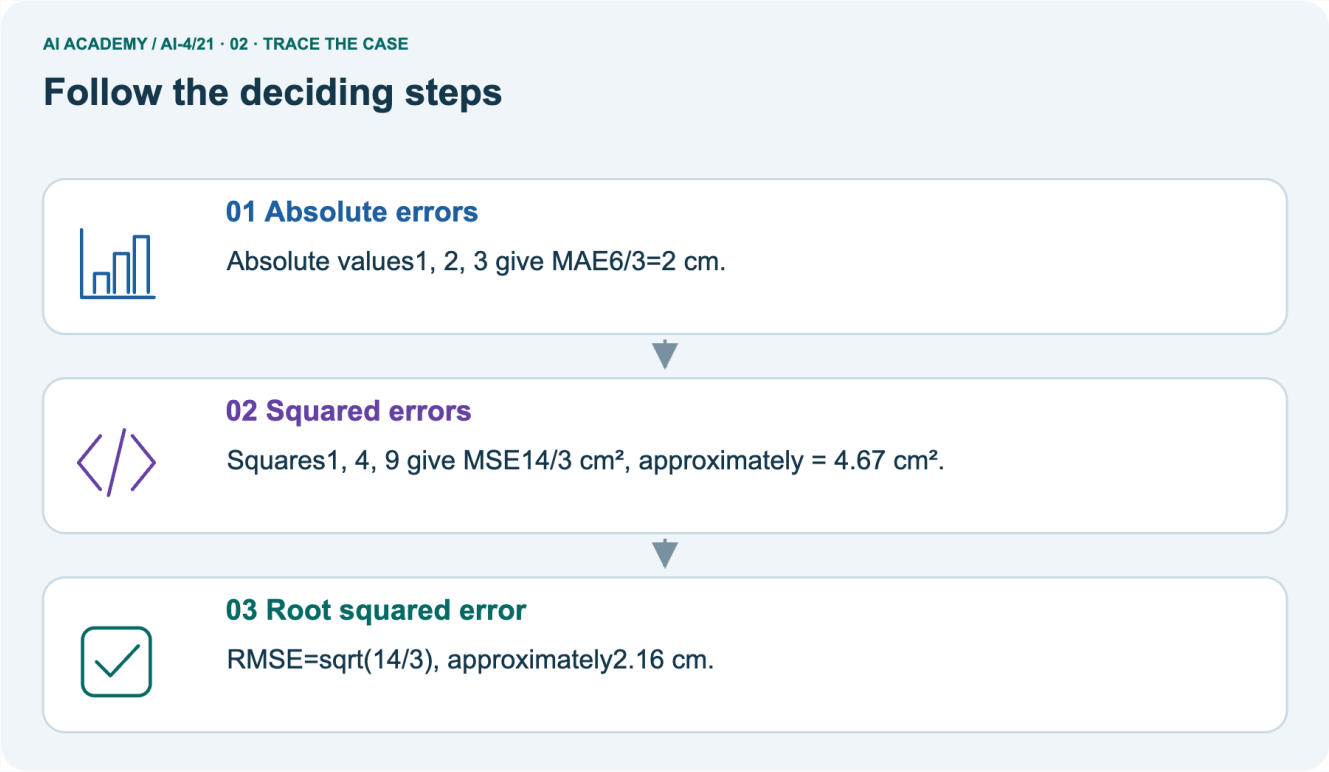
Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main residuals are [-1,2,-3] cm; the pictured residuals are [-1,2,3] cm. Their signs differ but their absolute and squared errors agree: MAE 2 cm, MSE 14/3 cm², RMSE sqrt(14/3)≈2.16 cm. Converting the pictured residuals to mm gives [-10,20,30]: MAE 20 mm, MSE 1400/3 mm²,

RMSE≈21.60 mm. The physical errors did not change. Never compare the raw cm and mm loss numbers as if the model improved or worsened.

Trace absolute, squared and rooted summaries



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Absolute errors	Absolute values1, 2, 3 give MAE6/3=2 cm.
2. Squared errors	Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm².
3. Root squared error	RMSE=sqrt(14/3), approximately2.16 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict converting every residual to millimetres

- Change one thing: Convert every residual to millimetres. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-4/21 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

RMSE=sqrt(14/3), approximately2.16 cm.

Changed condition

Convert every residual to millimetres.

Worked changed-case result

MAE becomes20 mm; MSE becomes1400/3 mm²; RMSE becomes about21.60 mm.
Why: Residuals scale by10; squares scale by100, while the square root returns a length unit.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the unit and consequence limit

- Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

- Read the labelled picture: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main MSE/RMSE question

Absolute and squared loss at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Residuals	minus 1 plus 2 minus 3	Signed differences
Absolute values	1 2 3	MAE 2 cm
Squares	1 4 9	MSE 14 divided by 3

Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

Give MSE and RMSE with units. Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 21 • WORKED EVIDENCE

1. Actual heights: 10, 14, 20; predictions: 12, 13, 17.
2. Residuals: −2, +1, +3 cm; absolute errors: 2, 1, 3 cm.
3. MAE = (2 + 1 + 3) ÷ 3 = 2 cm.
4. Compare with a baseline on the same cases and inspect the largest error.

STEP 1 Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17.

STEP 2 Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

STEP 3 MAE = (2 + 1 + 3) ÷ 3 = 2 cm.

STEP 4 Compare with a baseline on the same cases and inspect the largest error.

CHECK Verify the denominator, **unit**, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

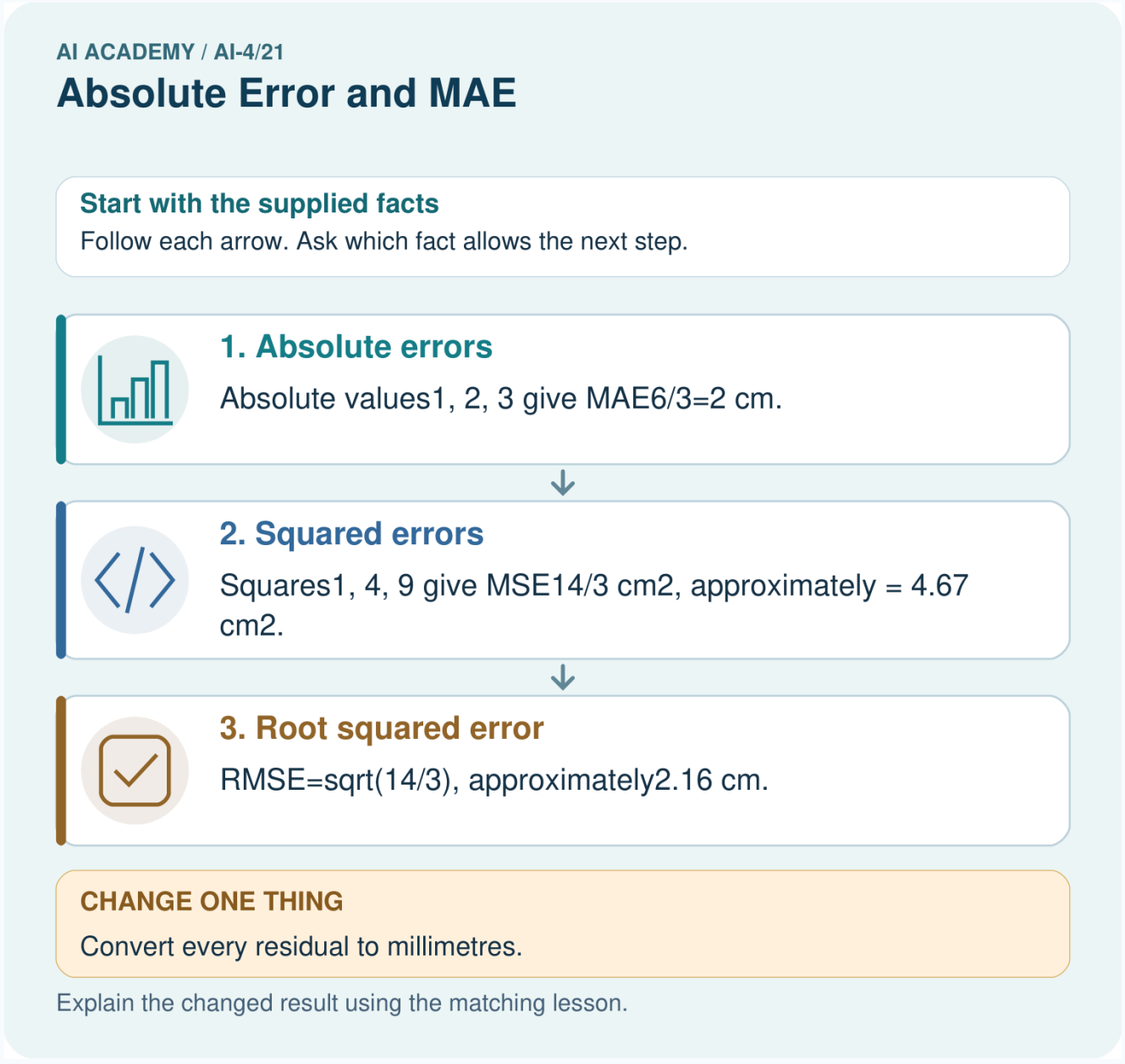
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Absolute Error and MAE

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a signed mean or unit change mistaken for lower loss

> Inspect the per-case residuals and the actual formula.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Average signed residual is a safe replacement for average **absolute error**.

Explain the interpretation and next check

- Choose and document the comparison metric before reviewing final-test results. Neither MAE nor MSE reveals all subgroup patterns or proves safety. Do not remove a large valid error solely because it worsens squared **loss**.

> **A team switches from MSE to MAE only after seeing a bad large error. What should be reported?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Average signed residual is a safe replacement for average absolute error. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise the same loss under a declared unit conversion

> MAE and RMSE scale by10 from cm to mm; MSE scales by100.

Try the supported squared-loss task

Use workbook W2 for the connected example “Absolute and squared **loss**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Explain the conversion and comparison conditions

Changing cm to mm multiplies residuals and MAE/**RMSE** by 10, but MSE by 100. A **loss** decrease across reports is not meaningful unless **unit**s, evaluated rows and **loss** definitions match.

Retain the per-record residuals for investigation.

Show evidence beyond a final number

- Use the worked case for Absolute and squared **loss**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured squared-unit error

- Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the independent metre-to-centimetre task

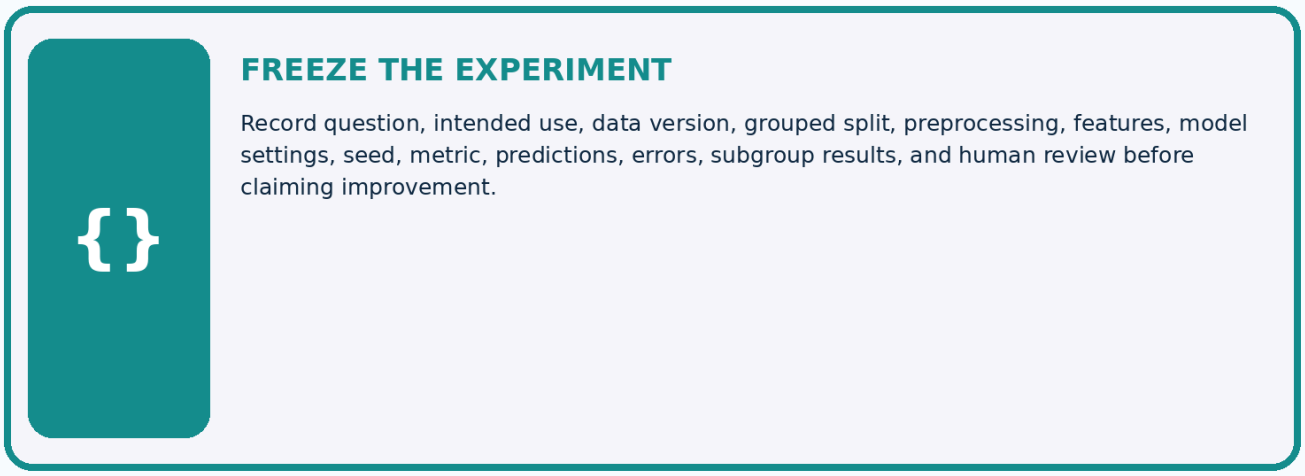
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + **unit**s + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Compute squared loss

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete same-case model and baseline loss comparison

> Every training target, prediction and evaluation outcome is supplied.

Complete matched E1–E3 evidence

Matched fictional evaluation ID	Observed height cm	Supplied candidate prediction cm	Training-mean baseline prediction cm
E1	10	10	7
E2	10	10	7
E3	10	4	7

Training-only baseline and checked loss rankings

Fit the mean baseline on separate training heights [5,7,9] cm, giving constant 7 cm; those targets are not evaluation cases. Candidate predictions [10,10,4] are supplied, not fitted in this lesson. On matched E1–E3, candidate residuals [0,0,6] give MAE 2 cm, MSE 12 cm², RMSE sqrt(12)≈3.46 cm. Baseline residuals [3,3,3] give MAE 3 cm, MSE 9 cm², RMSE 3 cm. The candidate is better by MAE but worse by MSE/RMSE here.

Explain the single large miss and state which loss was chosen for the task before scoring.

Normal, exact-zero, missing and unit-conversion probes

Complete loss probe	Expected bounded check
Normal residuals [-1,2,-3] cm	Absolute errors [1,2,3], squares [1,4,9]; MAE 2, MSE 14/3, RMSE sqrt(14/3).
Boundary residuals [0,0] cm	All three losses are zero on these supplied records, not proof of future exact predictions.
Missing residual list or unknown observation	No observed average loss; report unavailable/request the missing evidence rather than invent zero.
Changed unit from cm to mm for all matched errors	MAE/RMSE ×10, MSE ×100; unchanged physical errors and case identities.

Calculate each row and compare like with like

1. MAE Lab: list each candidate and baseline residual for E1–E3, then compute absolute errors, squares, totals and denominators. Compare each method using the same loss and unit before discussing why the loss choice reverses the ranking. Retain the largest candidate miss and do not remove it to improve the score. Predict the optional program outputs and record traced versus executed evidence.

   No final test or actual consequence weights are supplied. Keep the selected metre-residual task for an uncoached first attempt.

Optional complete guarded loss program

```
import math


def losses(residuals):
    if not residuals:
        raise ValueError('No observed average loss without records')
    if any(type(r) not in (int, float) or not math.isfinite(r) for r in residuals):
        raise ValueError('Need finite matched residuals in one declared unit')
    mae = sum(abs(r) for r in residuals) / len(residuals)
```

```
    mse = sum(r * r for r in residuals) / len(residuals)
    return {'mae': mae, 'mse': mse, 'rmse': math.sqrt(mse)}


training_heights = [5, 7, 9]  # Fit the constant before evaluating new cases.
baseline = sum(training_heights) / len(training_heights)
observed = [10, 10, 10]      # E1, E2, E3, three different fictional objects.
candidate = [10, 10, 4]     # Supplied predictions; candidate fitting not given.
baseline_predictions = [baseline] * len(observed)
for name, predictions in [('candidate', candidate), ('mean baseline',
baseline_predictions)]:
    residuals = [y - p for y, p in zip(observed, predictions)]
    print(name, residuals, losses(residuals))
# Length/order/unit identity is declared by the supplied matched E1–E3 fixture.
# Different losses can rank the same fixed predictions differently.
```

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

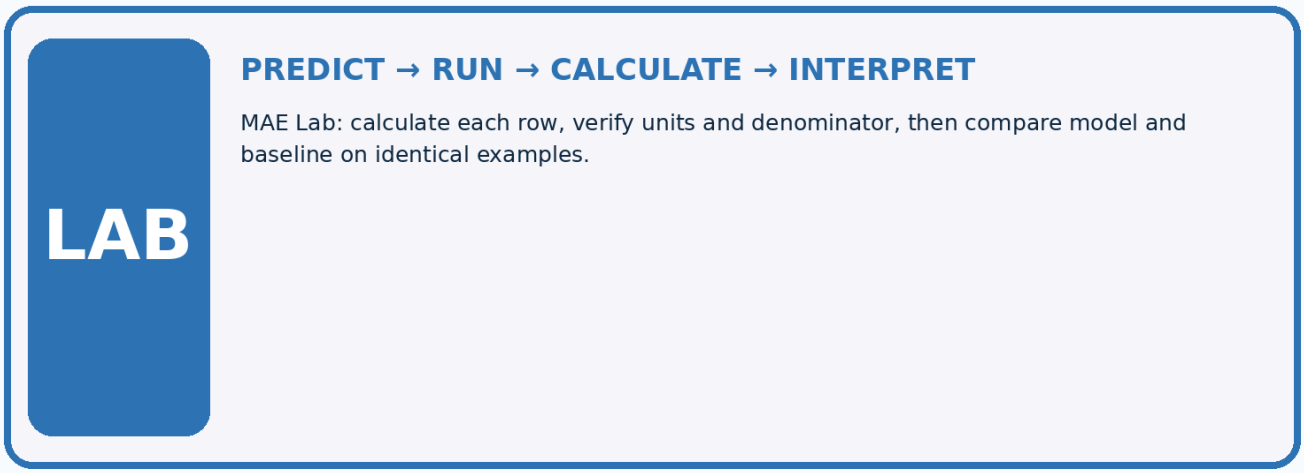
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. MAE Lab: calculate each row, verify **unit**s and denominator, then compare model and baseline on identical examples.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compute and convert losses from new residuals

> Keep the operation order and the unit conversion explicit.

Complete independent residuals and scope

Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

This independent task starts with residuals in metres. Apply each operation to the supplied three residuals before converting the final losses: metre-to-centimetre scale100 for MAE/RMSE and10000 for MSE. The independent values differ from the centimetre worked example. State squared units for MSE and do not infer new model predictions or unseen-case reliability.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply independently

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Actual heights are 4, 8 and 10 cm; **prediction**s are 5, 6 and 13 cm. Calculate **absolute error**s and MAE.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result
</details>

Write or draw your reasoning and evidence.

Connect to your project

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Keep the loss definition with the numeric experiment

> A result needs the same target,records and units as its comparison.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

For the numeric side model,save target quantity/time/unit,matching source IDs,per-case residuals,loss definition,denominator and the same-case baseline. Select the intended loss and consequence priorities before choosing the most flattering result. Keep classification metrics separate; a centimetre MAE is not an accuracy percentage.

Record missing outcomes and extreme misses rather than silently changing the evaluation set.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with the formula,unit and comparison contract

> Explain why loss ranking can depend on how large misses are weighted.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Calculate MAE MSE and **RMSE**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
absolute error	absolute value of actual minus predicted	
MAE	mean of absolute errors	
unit	measurement scale retained by MAE	
cancellation	positive and negative errors offset	

Closer-explanation terms

Word	Meaning
MAE	Mean absolute error
MSE	Mean squared error
RMSE	Square root of mean squared error

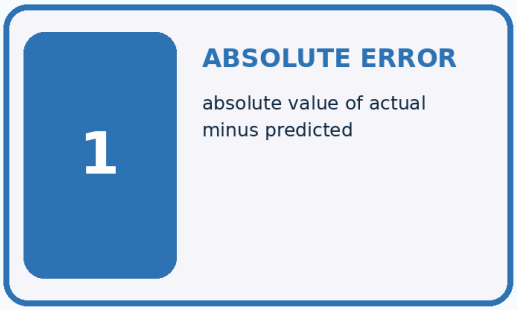
Word	Meaning
Loss	A numerical penalty for prediction errors

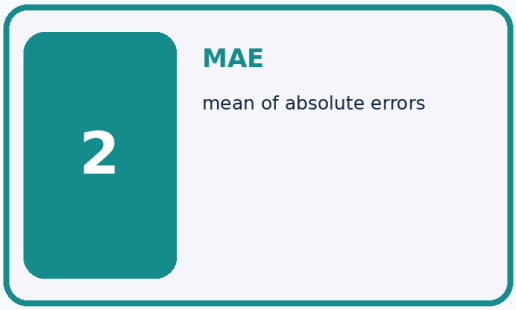
▼ Earlier wording, alternatives and picture transcripts

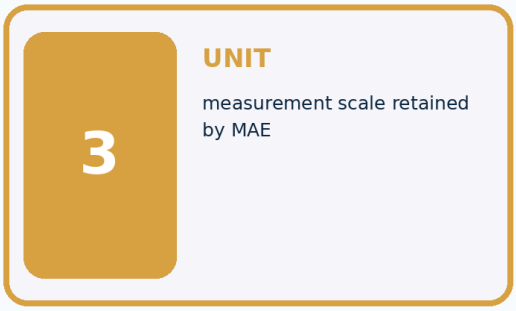
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Average signed residual is a safe replacement for average **absolute error**.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **absolute error**, MAE, **unit**, **cancellation**.
- CALCULATE Reproduce the worked **evidence** with **unit**s and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Average signed residual is a safe replacement for average **absolute error**.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

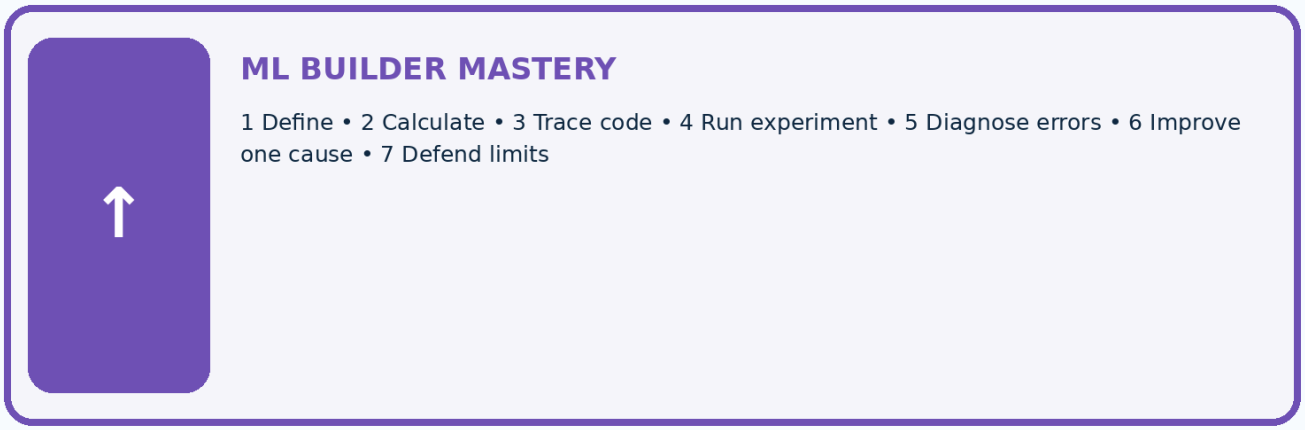
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh metre-residual and centimetre-conversion task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

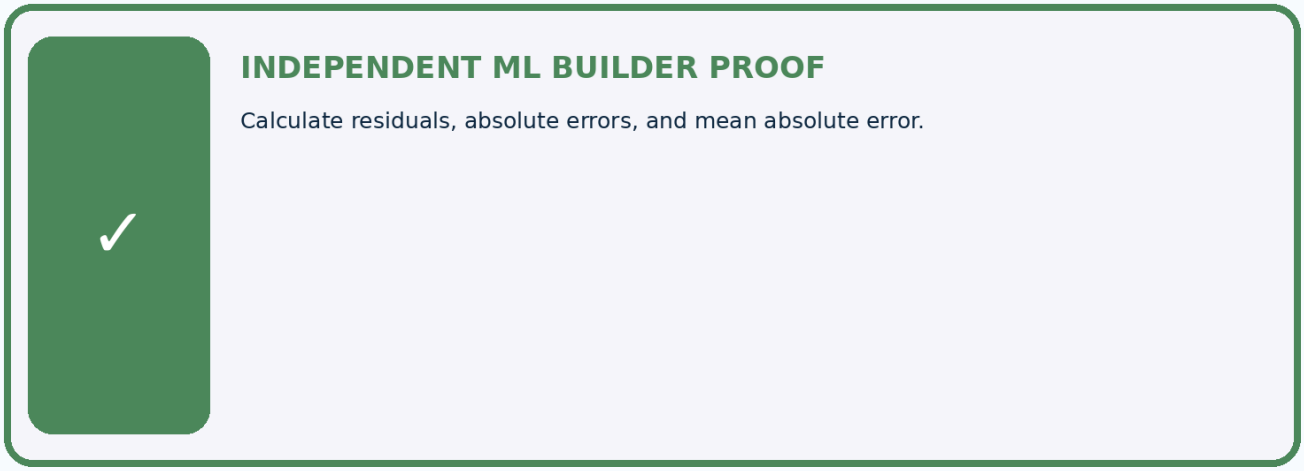
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Calculate residuals, **absolute error**s, and mean **absolute error**.

NEXT BRIDGE Carry the experiment **evidence** into: Train a Regression Model.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Evidence or stage	Value or result
Case 1, actual 3, predicted 2	1
Case 2, actual 5, predicted 7	2
Case 3, actual 9, predicted 6	3

▼ Earlier wording, alternatives and picture transcripts

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Absolute errors are measured before averaging

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Calculate MAE MSE and RMSE	Explain the deciding step, show a trace or test, and state one limit.
Explain how squaring changes sensitivity to large errors	Explain the deciding step, show a trace or test, and state one limit.
Match the loss and units to the task	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Compare absolute and squared error without losing track of **unit**s.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Compare absolute and squared error without losing track of **unit**s.

- Which labelled step matches this detail? **RMSE**=sqrt(14/3), approximately2.16 cm.
- 1. **Absolute error**s
- 2. Root squared error
- 3. Squared errors
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Root squared error **RMSE**=sqrt(14/3), approximately2.16 cm. Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Convert every residual to millimetres. Which conclusion is supported by the supplied facts?
- 1. MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm. Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Repair and keep the record

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
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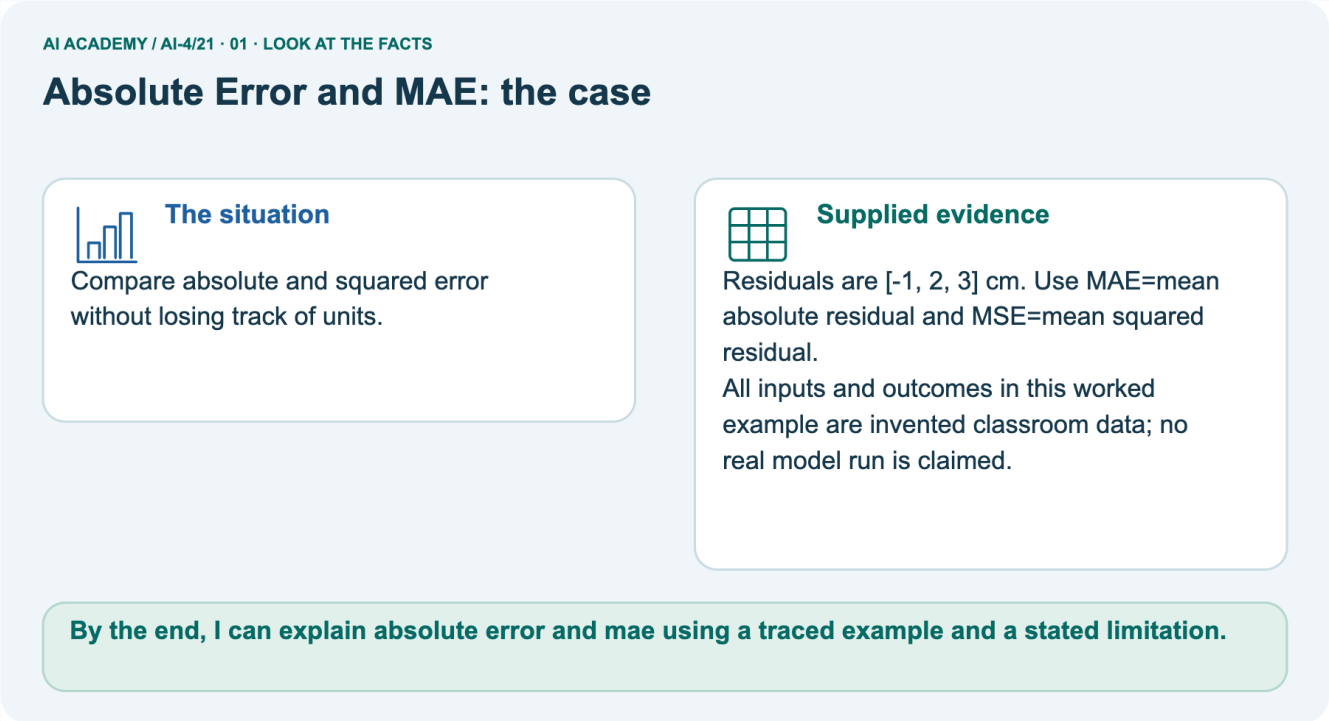
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Absolute Error and MAE: the case



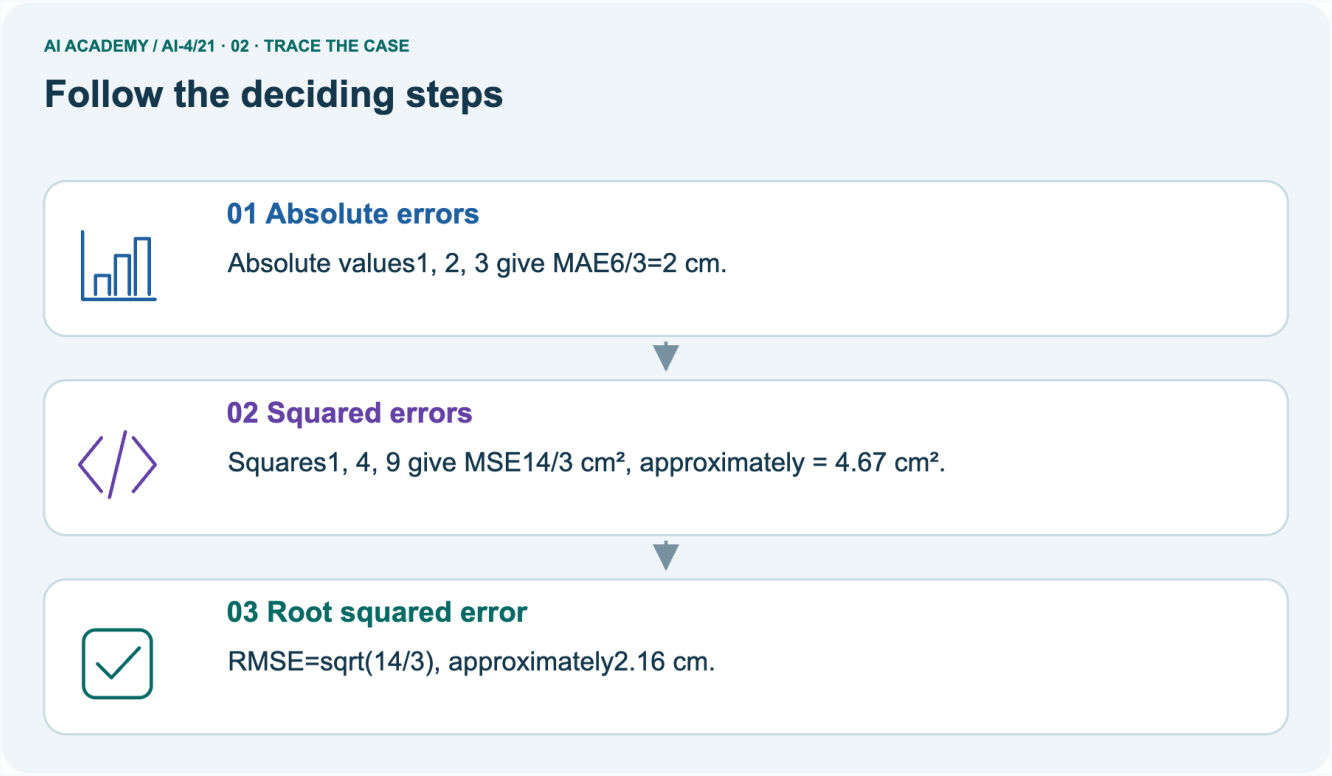
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare absolute and squared error without losing track of **unit**s. Supplied facts: Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute errors: Absolute values1, 2, 3 give MAE6/3=2 cm. → Squared errors: Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm². → Root squared error: **RMSE**=sqrt(14/3), approximately2.16 cm.

Point to each numbered stage and explain which recorded **input** it uses.

03. Absolute and squared loss at a glance

AI ACADEMY / AI-4/21 · 03 · INSPECT THE LESSON MODEL

Absolute and squared loss at a glance

Part	Evidence or action	Interpretation
Residuals	minus 1 plus 2 minus 3	Signed differences
Absolute values	1 2 3	MAE 2 cm
Squares	1 4 9	MSE 14 divided by 3

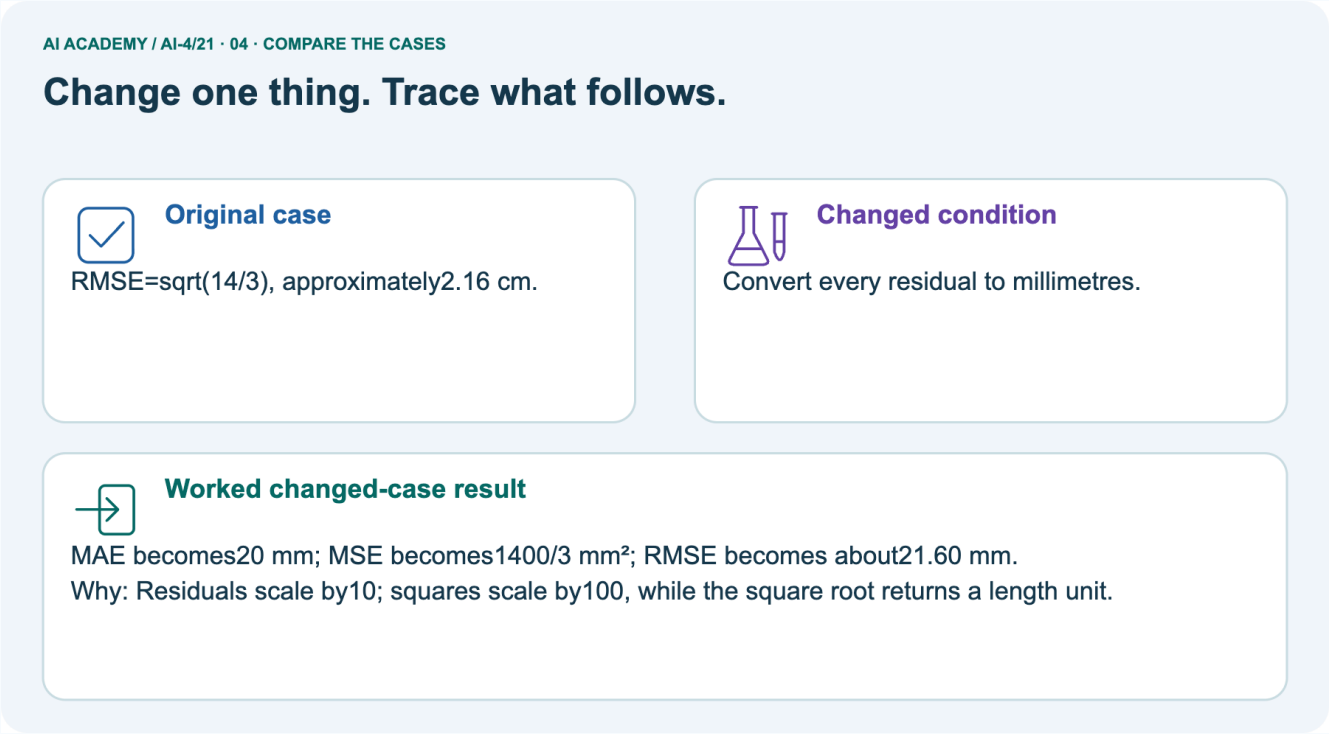
Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute and squared **loss** at a glance. Residuals | minus 1 plus 2 minus 3 | Signed differences. Absolute values | 1 2 3 | MAE 2 cm. Squares | 1 4 9 | MSE 14 divided by 3

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



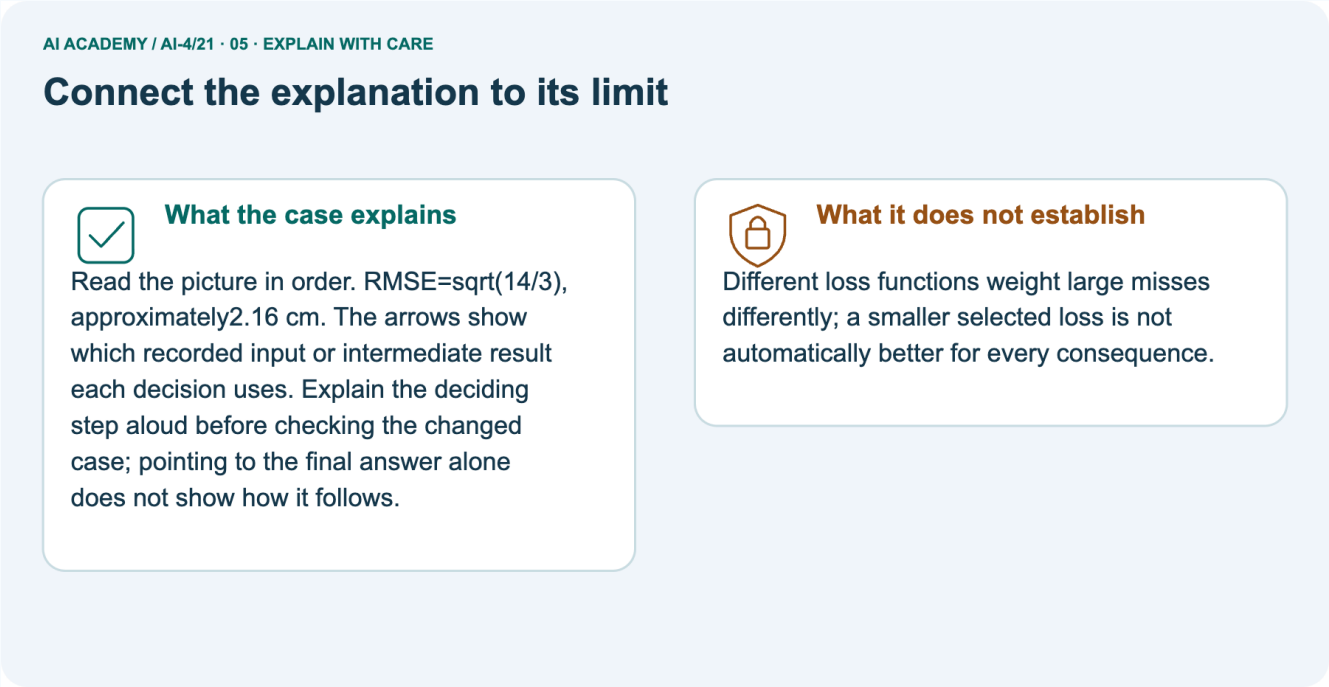
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **RMSE**=sqrt(14/3), approximately2.16 cm. Changed condition: Convert every residual to millimetres. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

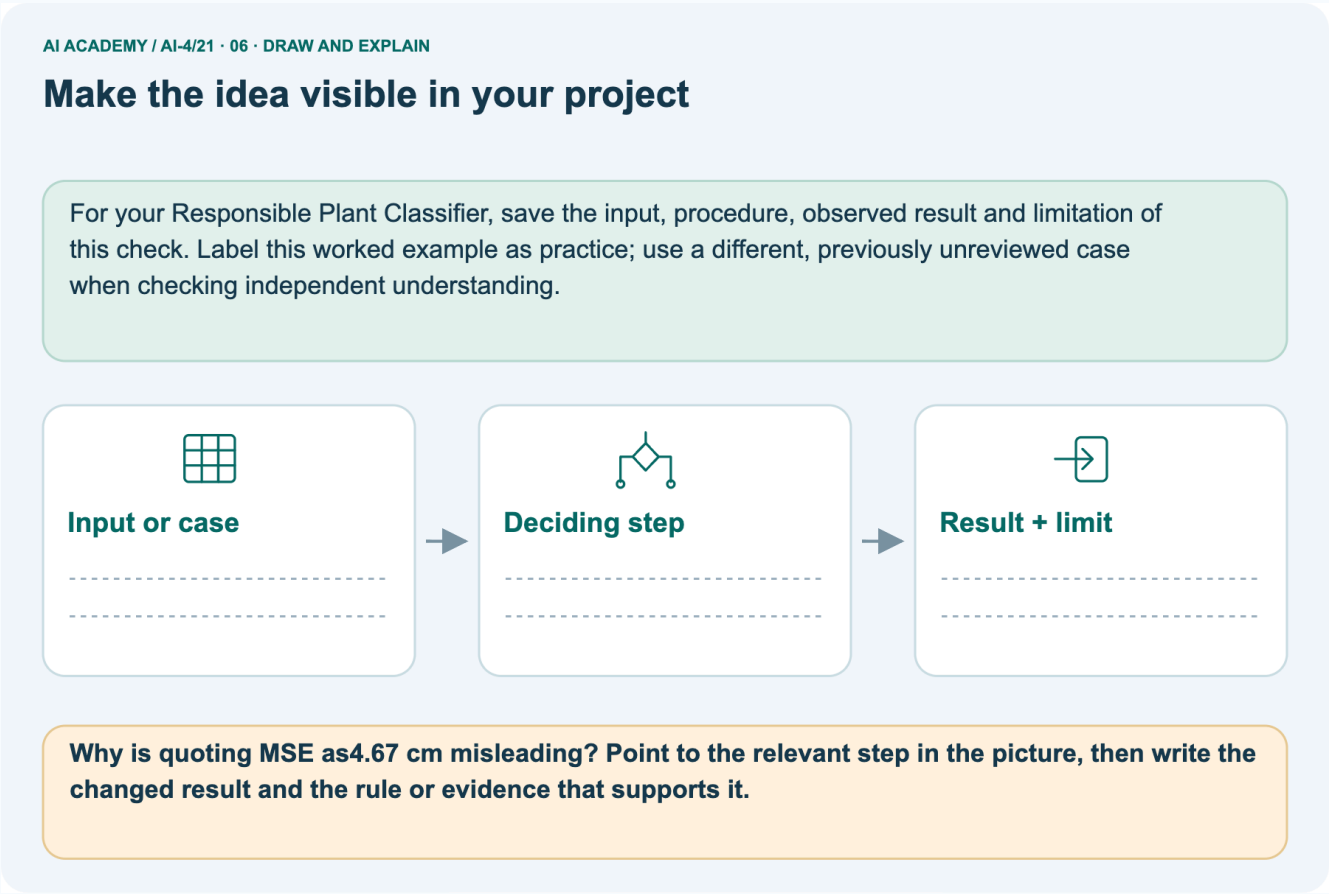
Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Absolute Error and MAE: the case

02. Follow the deciding steps

03. Absolute and squared loss at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.